FINAL LABORATORY REPORT

Microsoft 365 and Active Directory IT Support and Identity Lifecycle

Configuration, validation, troubleshooting, and user offboarding

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Environment: Windows Server, Windows 11, and Microsoft 365

A laboratory study of identity and access administration in a Windows domain and a separate Microsoft 365 tenant.

Abstract

This project implemented and tested an identity-management laboratory using Active Directory Domain Services (AD DS), Windows 11, and Microsoft 365. DC02, a Windows Server 2025 virtual machine, provided directory and DNS services for the support.test domain. WIN11-01 served as the domain workstation. A standard account, Alex Support, was used to test domain sign-in, Group Policy, password support, and Remote Desktop authorization. A separate cloud account was provisioned for Microsoft 365 licensing, Outlook access, authentication-method registration, sign-in review, and SharePoint permissions.

The laboratory demonstrated successful domain service discovery, workstation policy application, password-support workflows, corrected Remote Desktop authorization, and controlled SharePoint access. Offboarding tests showed rejection of the disabled AD account and blocked cloud account, together with session-revocation confirmation and removal of site visitor membership. Troubleshooting addressed client addressing, DNS selection, group permissions, and incorrect browser identity. The report distinguishes configured security controls from observed outcomes: Authenticator registration and Security Defaults were confirmed, but the inspected authentication event did not establish an MFA challenge. Selected technical screenshots support the findings.

Keywords: Active Directory; DNS; Group Policy; Microsoft 365; identity lifecycle; IT support.

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1. Introduction and objectives

IT support staff administer identities throughout their lifecycle: provisioning accounts, assigning permissions, resolving access problems, and removing access when it is no longer required. A valid account can still fail to reach a service when DNS, computer membership, policy scope, or authorization is incorrect. This laboratory examined these dependencies through a controlled test environment.

The aim was to configure a small Windows domain and a separate cloud tenant, validate a standard user’s access, and document the corrective actions taken when a test failed. The local and cloud directories were administered independently; no directory synchronization was configured.

1.1 Objectives

1. Establish AD DS and DNS and verify domain-user access on a Windows client.

2. Organize directory objects and apply a workstation security policy.

3. Perform password support and correct a Remote Desktop authorization failure.

4. Provision a cloud user and validate licensed Microsoft 365 access.

5. Review authentication settings and demonstrate SharePoint permission changes.

6. Offboard the test user and validate the resulting access restrictions.

1.2 Scope

The work used virtual machines and test identities. It covered local directory administration, selected cloud services, and user offboarding. Production deployment, hybrid identity synchronization, Intune management, custom Conditional Access, and email-delivery testing were outside the scope.

2. Laboratory environment and design

Oracle VirtualBox hosted the Windows domain environment. A shared NAT Network allowed the server and workstation to communicate and provided outbound connectivity. Unlike separate default NAT adapters, this mode placed the VMs on a common virtual network [1].

Table 1. Environment configuration.

Component	Recorded configuration
Domain controller	DC02; Windows Server 2025 Standard Evaluation with Desktop Experience; 4698 MB RAM, four virtual processors, 60 GB disk.
Workstation	WIN11-01; Windows 11 domain member.
Network	10.77.0.0/24; gateway 10.77.0.1; DC02: 10.77.0.10; client: 10.77.0.20.
DNS	Domain clients use 10.77.0.10. DC02 forwards unresolved queries to 1.1.1.1, with root-hints fallback enabled.
AD directory	Domain: support.test; NetBIOS name: SUPPORT; OUs: LabUsers, LabComputers, LabGroups.
Local identity	SUPPORT\alex.support; domain security group RDP-Lab-Users.
Cloud tenant	soranutitsupportlab.onmicrosoft.com; administrator labadmin and standard user alex.support at that suffix.
Cloud services	Microsoft 365, Microsoft Entra ID, Outlook, Microsoft Authenticator, and SharePoint Online.

The matching Alex display names represented separate accounts. Local password or account-state changes did not automatically change the cloud account. This design required independent support and offboarding actions in both directories.

3. Methodology

3.1 Build and validation sequence

The environment was built in stages. Server addressing and DNS were established before the client joined the domain. Directory objects and group membership were then configured, followed by workstation policy and support tests. Cloud provisioning and resource-access tests were completed before offboarding.

Each test compared the intended result with the visible output. Positive tests demonstrated access after provisioning or correction. Negative tests demonstrated rejected access before a permission grant or after account restriction. The signed-in user and target machine were checked so that an administrator session would not be mistaken for a standard-user result.

3.2 Validation criteria

Test area	Acceptance criterion
DNS and membership	Host and LDAP service records identify DC02; the workstation opens the intended domain-user session.
Group Policy	Resultant policy lists LAB-Workstation-Baseline on WIN11-01.
Password support	The user completes the required change and regains the intended session or application access.
Authorization	The intended user reaches the service after the relevant group or resource permission is assigned.
Offboarding	Administrative account restrictions are followed by fresh rejected sign-ins where captured; resource membership removal is confirmed.

3.3 Evidence selection

The evidence appendix contains 29 selected screenshots covering the principal configuration and validation results. Repeated screens and routine intermediate navigation were omitted. Figure captions distinguish confirmed outcomes from limitations and retain the original exhibit IDs (E01-E86) used by the companion walkthrough. The original screenshots were retained separately; sensitive authentication-enrollment material is not reproduced.

A selected setting was not treated as proof that a change had been saved. Administrative success messages and user-side access tests were evaluated separately. Screenshot timestamps were retained as displayed and were not converted into a forensic timeline.

4. Implementation

4.1 Domain infrastructure and workstation

DC02 received the static address 10.77.0.10/24 and was promoted into the new support.test forest with AD DS and DNS. The server appeared as a global catalog domain controller. Its DNS client pointed to the local DNS service, and forwarding was configured for external names. Host, LDAP SRV, and public-name lookups returned answers (Figures 1, 2, 3, 4) [2-4].

LabUsers, LabComputers, and LabGroups organized the directory objects. Alex Support was created as a standard user. The client's duplicate address and incorrect mask were corrected to 10.77.0.20/24 before domain membership was validated. The workstation was placed in LabComputers, and a session identified Alex on WIN11-01 (Figures 5, 6).

4.2 Policy and support tasks

LAB-Workstation-Baseline configured a 600-second machine inactivity limit and appeared in the client's applied computer policies (Figures 7, 8). The AD password-support test required the user to set a new password, followed by a successful session (Figures 9, 10). Remote Desktop access was corrected by connecting Alex's domain group to the workstation's local Remote Desktop Users group (Figures 11, 12, 13, 14) [5].

4.3 Cloud provisioning and resource access

The cloud Alex account received the product license and opened Outlook. Authenticator registration and enabled Security Defaults were captured. Sign-in details were reviewed, and a cloud password reset was followed by restored Outlook access (Figures 15, 16, 17, 18, 19, 20, 21) [6-8].

A SharePoint communication site named IT Support Knowledge Base was created. Alex was initially denied access, then added as a site visitor. The direct site URL subsequently loaded in the test-user context (Figures 22, 23) [9].

4.4 Offboarding

Alex's AD account was disabled. Cloud sign-in was blocked, sign-in sessions were revoked, and the SharePoint visitor membership was removed. Fresh sign-in attempts demonstrated local and cloud rejection (Figures 24, 25, 26, 27, 28, 29) [10,11].

5. Results

Table 2. Principal validation results.

Test	Observed result	Figures
DNS discovery	DC02 host and LDAP service records resolved through the internal DNS server.	4, 5
Domain session	whoami and hostname identified Alex on WIN11-01.	6
Workstation policy	The 600-second setting and applied baseline GPO were visible.	7, 8
AD password reset	The password-change form was followed by a verified Alex session.	9, 10
Remote Desktop	An authorization error was corrected; domain and local group membership supported the successful retest.	11-14
Cloud provisioning	Alex was licensed and Outlook opened.	15, 16
Authentication setup	Authenticator registration and enabled Security Defaults were confirmed.	17, 18
Authentication event	The selected event recorded a successful password step; it did not establish an MFA challenge.	19
Cloud password reset	Reset confirmation was followed by Outlook access after the password-change workflow.	20, 21
SharePoint access	Initial denial was followed by successful site access after the visitor grant.	22, 23
Local offboarding	Disabled account state and rejected sign-in were captured.	24, 25
Cloud offboarding	Sign-in block, revocation confirmation, rejected sign-in, and removal of site visitor membership were captured.	26-29

6. Troubleshooting and corrective actions

6.1 Addressing and DNS

The client initially reused the server's 10.77.0.10 address and had a 255.0.0.0 mask. DNS queries returned no response. Assigning the unique 10.77.0.20 address with the intended /24 mask corrected the configuration, and subsequent host and service lookups succeeded (Figure 5). This demonstrated that a directory-access problem can originate in basic networking.

A public DNS resolver did not resolve the private dc02.support.test record. The internal DNS service resolved both the domain host and a public website. Domain members therefore used 10.77.0.10, with external resolution handled through the DNS server's forwarding configuration (Figures 3, 4) [3,4].

6.2 Remote Desktop authorization

The RDP error identified missing remote-login authorization. The correction used two group relationships: Alex belonged to RDP-Lab-Users, and that domain group belonged to the workstation's local Remote Desktop Users group. A later remote session confirmed the intended account and computer (Figures 11, 12, 13, 14). The result demonstrated that a valid password and an existing account do not by themselves grant remote access.

6.3 Browser identity and session controls

An administrator browser session initially confused the SharePoint access test. Repeating the test in Alex's context established the denial before a visitor grant and successful access afterward (Figures 22, 23). Checking the active identity was essential to interpreting the result.

During offboarding, the expected Sign out all sessions option was not visible in the Microsoft 365 pane. The Entra Revoke sessions action was used and returned a success confirmation (Figure 27). The confirmation established the submitted administrative action without assuming that every application token stopped immediately [11].

7. Discussion and limitations

The findings show the dependency between networking, DNS, authentication, and authorization. Correct addressing enabled communication; DNS identified directory services; authentication established identity; and group or resource permissions determined access. Inspecting these layers separately helped connect each failure to an appropriate correction.

The group-based RDP permission and SharePoint visitor assignment illustrated least privilege. Alex remained a standard user while receiving the access required for specific tests. Using an administrator account for those tests would have hidden missing permissions and weakened the evidence.

The authentication review also demonstrated the difference between configuration and observed behavior. Registering Authenticator and enabling Security Defaults did not establish that the inspected sign-in completed an MFA challenge. That event showed a successful password step only (Figure 19) [6,7].

7.1 Limitations

Area	Qualification
Inactivity control	The setting and applied GPO were captured. Exact elapsed lock time and a final post-restoration timing test were not independently recorded.
AD group removal	The final membership view showed Domain Users only, but Apply remained enabled. Persistence of that membership change was not established by a reopened post-save view.
Cloud services	Outlook access was verified; mail delivery was not. SharePoint site access was verified; a separate file-edit denial was not captured.
Offboarding scope	One cloud license and one group membership remained. The results concern specific account restrictions and site permission removal, not an exhaustive entitlement audit.
Evidence method	Screenshots show visible states and selected tests. They do not establish every background action or application-session outcome.

8. Conclusion

The project established a functioning AD DS/DNS environment and a separately administered Microsoft 365 tenant. The principal results were successful domain sign-in, workstation policy application, password-support workflows, corrected Remote Desktop authorization, licensed cloud access, SharePoint permission changes, and user offboarding.

The technical work demonstrated a repeatable support method: confirm the intended account and device, inspect the relevant dependency, make a scoped change, and validate the result from the user’s perspective. The account lifecycle required separate actions in the local and cloud directories because no synchronization was configured.

The report’s findings are limited to the captured tests. A specific MFA challenge, file-level permission denial, and a complete review of remaining entitlements would provide useful extensions. These were not counted as completed outcomes.

References

Technical documentation supporting the method and interpretation is listed below. Laboratory results are supported by the numbered figures in the evidence appendix.

[1] Oracle VirtualBox - Virtual Networking. Accessed 1 October 2026.

[2] Microsoft - Install Active Directory Domain Services. Accessed 1 October 2026.

[3] Microsoft - DNS client configuration best practices. Accessed 1 October 2026.

[4] Microsoft - DNS forwarding. Accessed 1 October 2026.

[5] Microsoft - Interactive logon: Machine inactivity limit. Accessed 1 October 2026.

[6] Microsoft - Security Defaults. Accessed 1 October 2026.

[7] Microsoft - Sign-in log activity details. Accessed 1 October 2026.

[8] Microsoft - Reset passwords. Accessed 1 October 2026.

[9] Microsoft - Manage site administrators and membership. Accessed 1 October 2026.

[10] Microsoft - Remove a former employee, step 1. Accessed 1 October 2026.

[11] Microsoft - Revoke user access. Accessed 1 October 2026.

Appendix A. Domain controller and DNS configuration

Figure 1. Domain controller and global catalog (E13). DC02 is listed in the support.test Domain Controllers OU as a global catalog (GC) server.

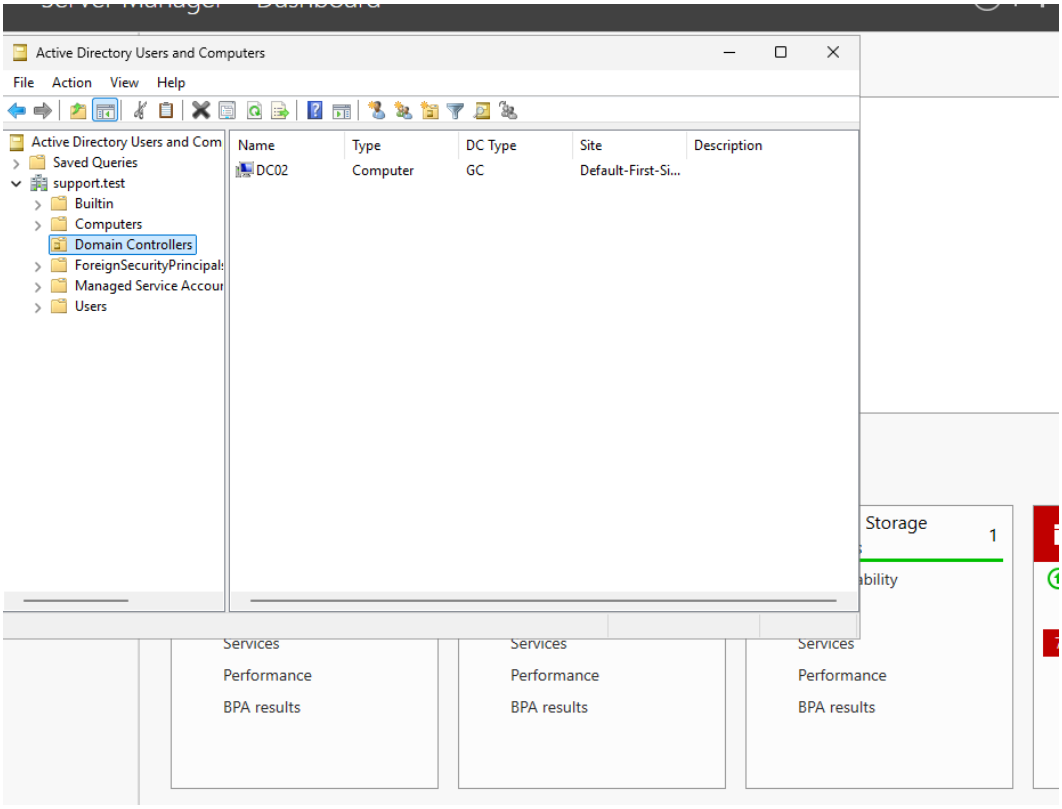
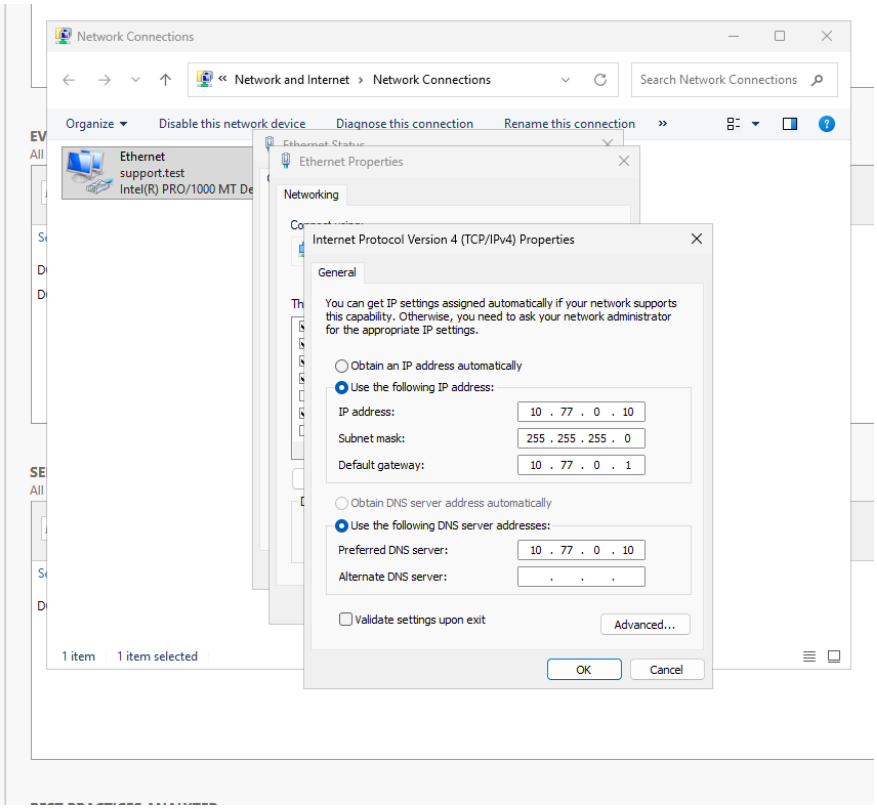


Figure 2. DC02 uses its own DNS service (E14). The server uses static address 10.77.0.10/24, gateway 10.77.0.1, and its own DNS service at 10.77.0.10.



Appendix A. DNS forwarding and service discovery

Figure 3. DNS zones and forwarder (E15). DNS Manager shows the support.test zone and upstream forwarder 1.1.1.1. Root-hints fallback is enabled.

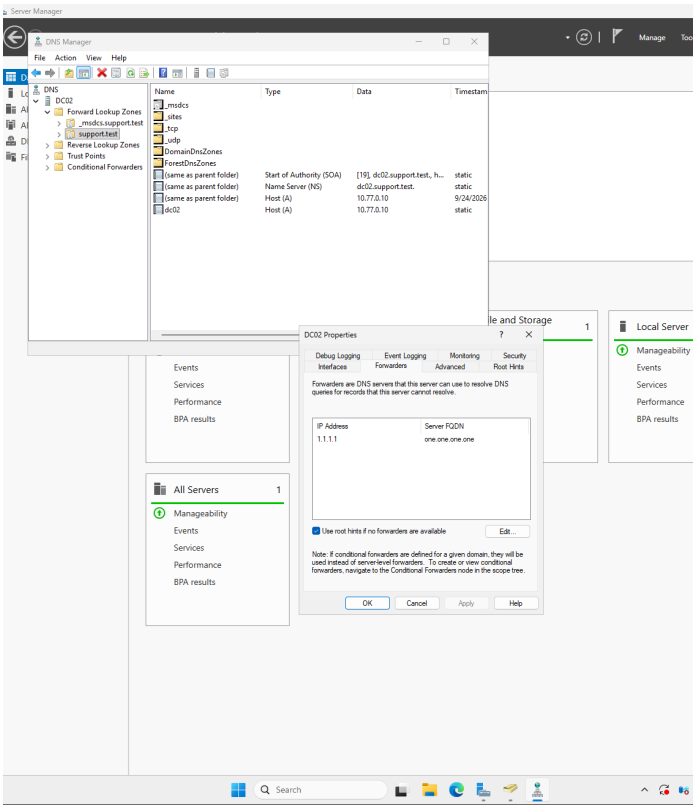


Figure 4. Internal, service, and external DNS checks (E16). Host, LDAP SRV, and public-name lookups return answers through 10.77.0.10. The SRV answer identifies DC02 on port 389.

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> nslookup dc02.support.test 10.77.0.10
Server:  UnKnown
Address:  10.77.0.10

Name:    dc02.support.test
Address:  10.77.0.10

PS C:\WINDOWS\system32> nslookup -type=SRV _ldap._tcp.dc._msdcs.support.test 10.77.0.10
Server:  UnKnown
Address:  10.77.0.10

_ldap._tcp.dc._msdcs.support.test       SRV service location:
          priority       = 0
          weight         = 100
          port           = 389
          svr hostname   = dc02.support.test
dc02.support.test       internet address = 10.77.0.10
PS C:\WINDOWS\system32> nslookup www.microsoft.com 10.77.0.10
Server:  UnKnown
Address:  10.77.0.10

Non-authoritative answer:
Name:    e13678.dscb.akamaiedge.net
Addresses:  2600:140a:1000:586::356e
          2600:140a:1000:58e::356e
          2600:140a:1000:58d::356e
          23.6.245.17
Aliases:  www.microsoft.com
          www.microsoft.com-c-3.edgekey.net

PS C:\WINDOWS\system32>
```

Appendix A. Client configuration and domain sign-in

Figure 5. Corrected client IP and domain service discovery (E22). The corrected client uses 10.77.0.20/24 and resolves the domain-controller host and service records.

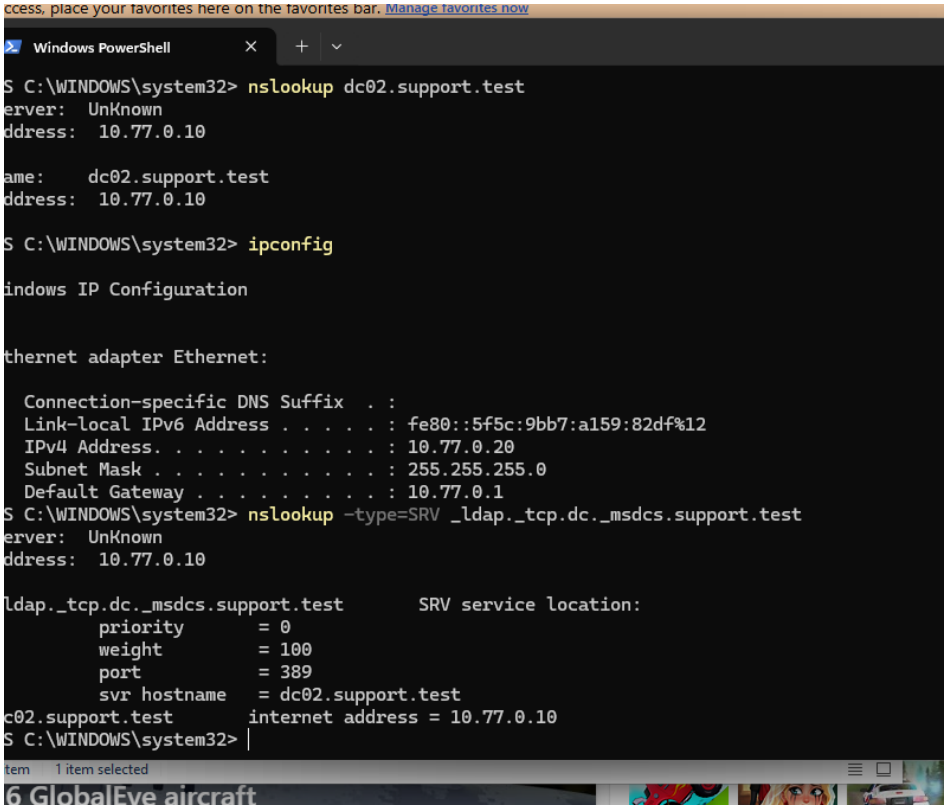
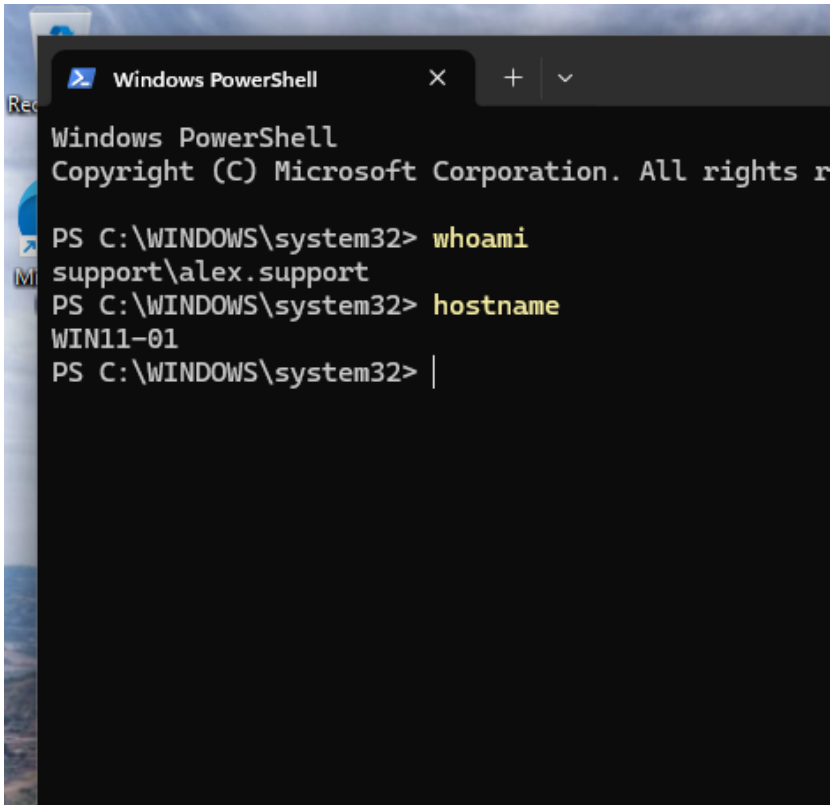


Figure 6. Domain sign-in verified (E24). The commands identify the signed-in user as support\alex.support and the workstation as WIN11-01.



Appendix A. Workstation policy and applied results

Figure 7. Workstation inactivity policy (E27). The workstation GPO defines Interactive logon: Machine inactivity limit as 600 seconds.

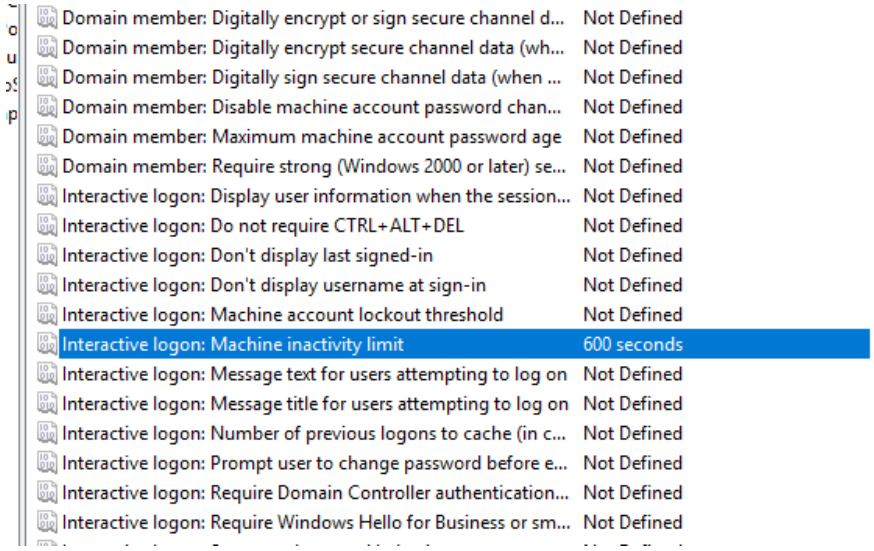
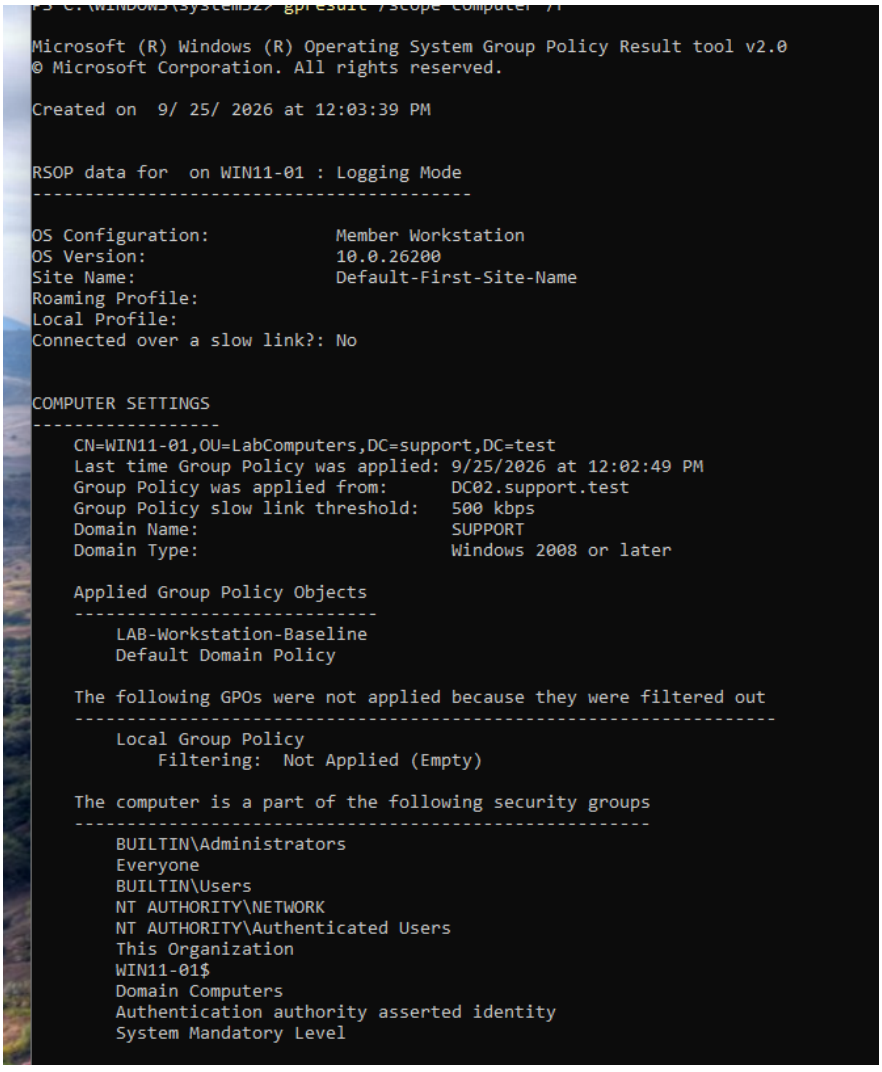


Figure 8. Applied Group Policy results (E28). The resultant computer-policy output lists LAB-Workstation-Baseline as applied to the workstation.



Appendix A. AD password support

Figure 9. AD password change form (E32). The domain user is prompted to set and confirm a new password during the password-support test.

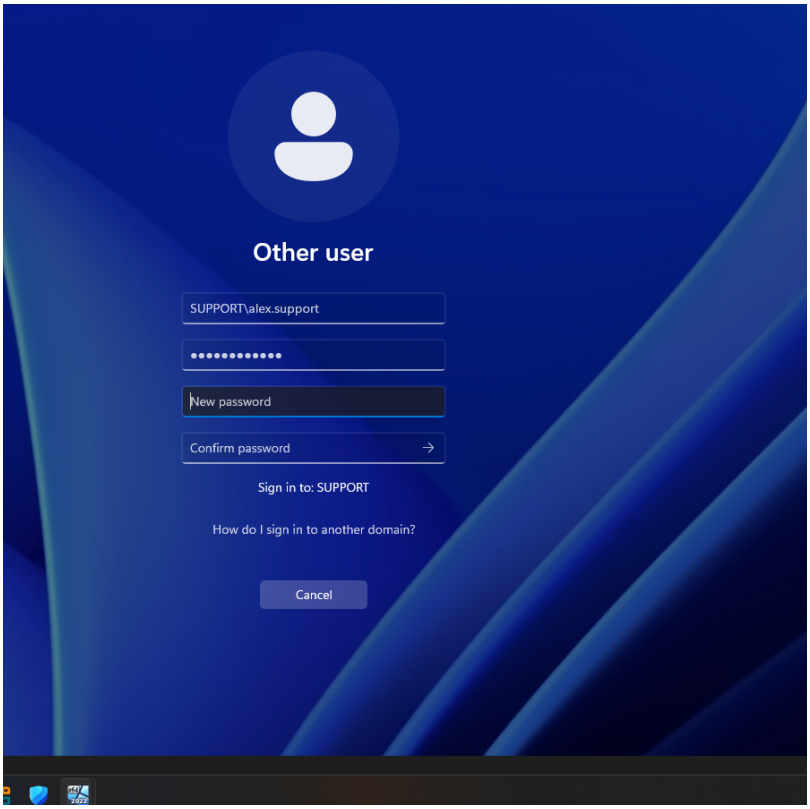
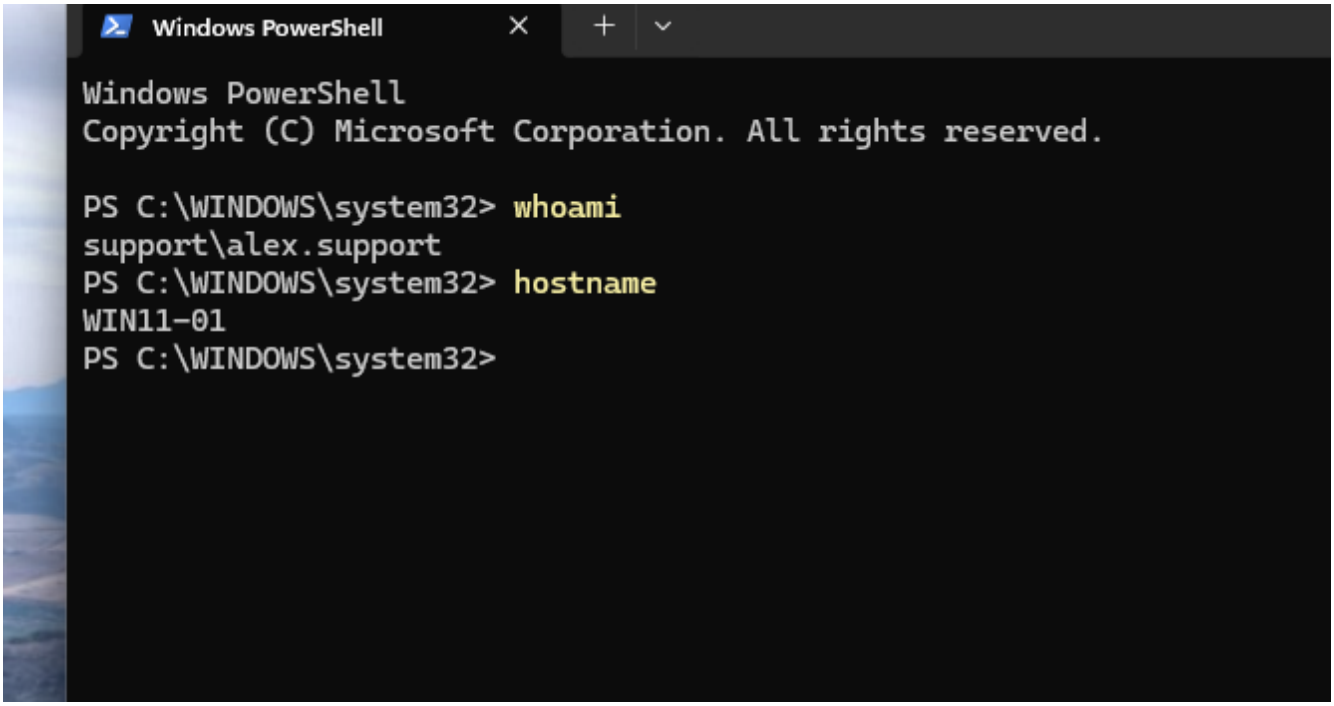


Figure 10. Successful sign-in after password task (E33). After the password task, whoami and hostname confirm Alex on WIN11-01.



Appendix A. Remote Desktop failure and successful retest

Figure 11. Remote Desktop authorization failure (E36). The initial Remote Desktop attempt fails because the user is not authorized for remote login.

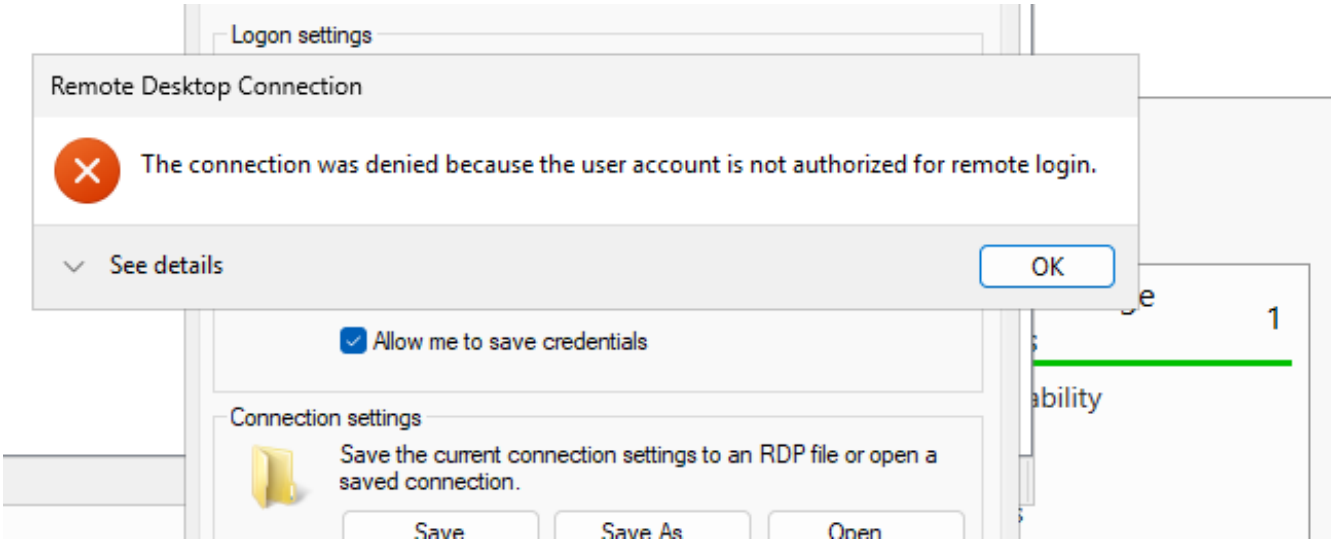
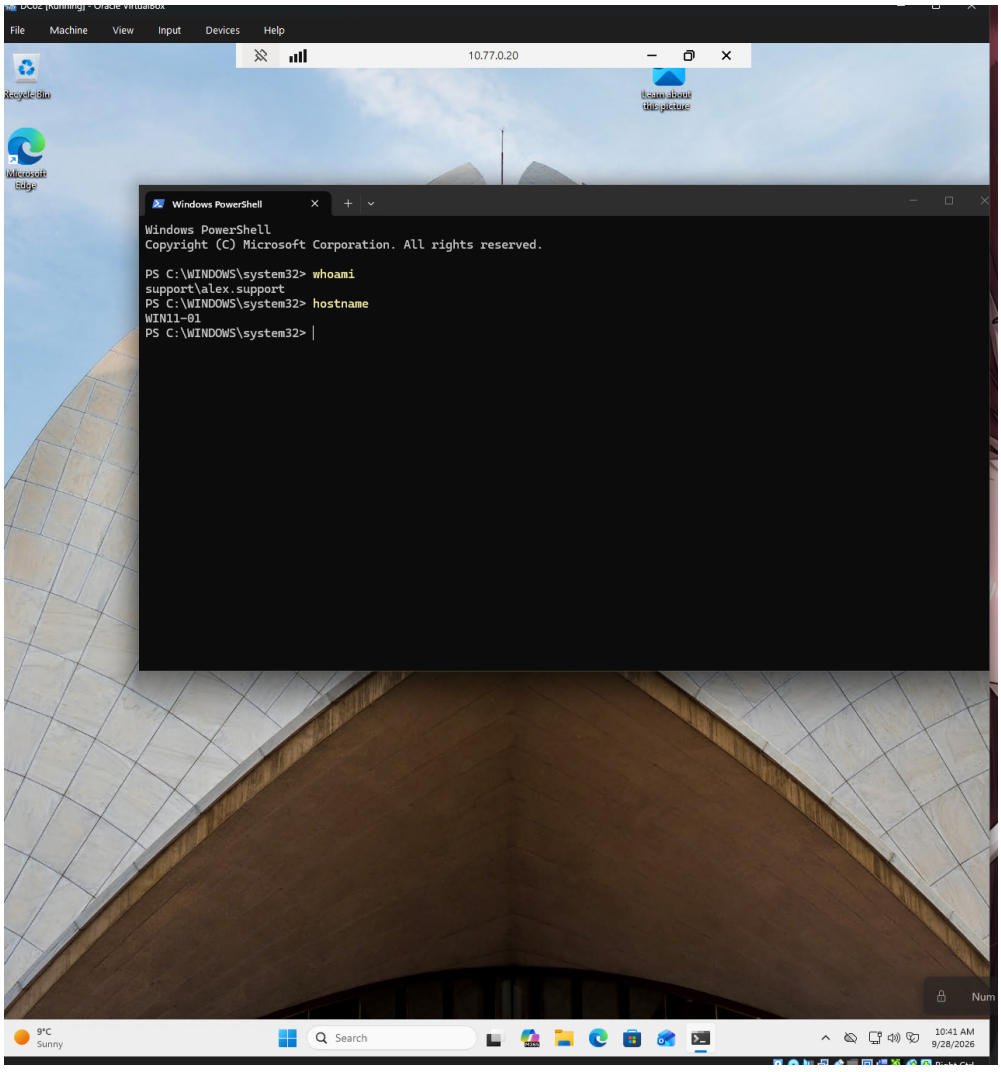


Figure 12. Remote Desktop session works (E39). The later remote session identifies support\alex.support on WIN11-01, validating the authorization correction.



Appendix A. Remote Desktop authorization groups

Figure 13. AD group membership (E40). Alex belongs to the domain security group RDP-Lab-Users.

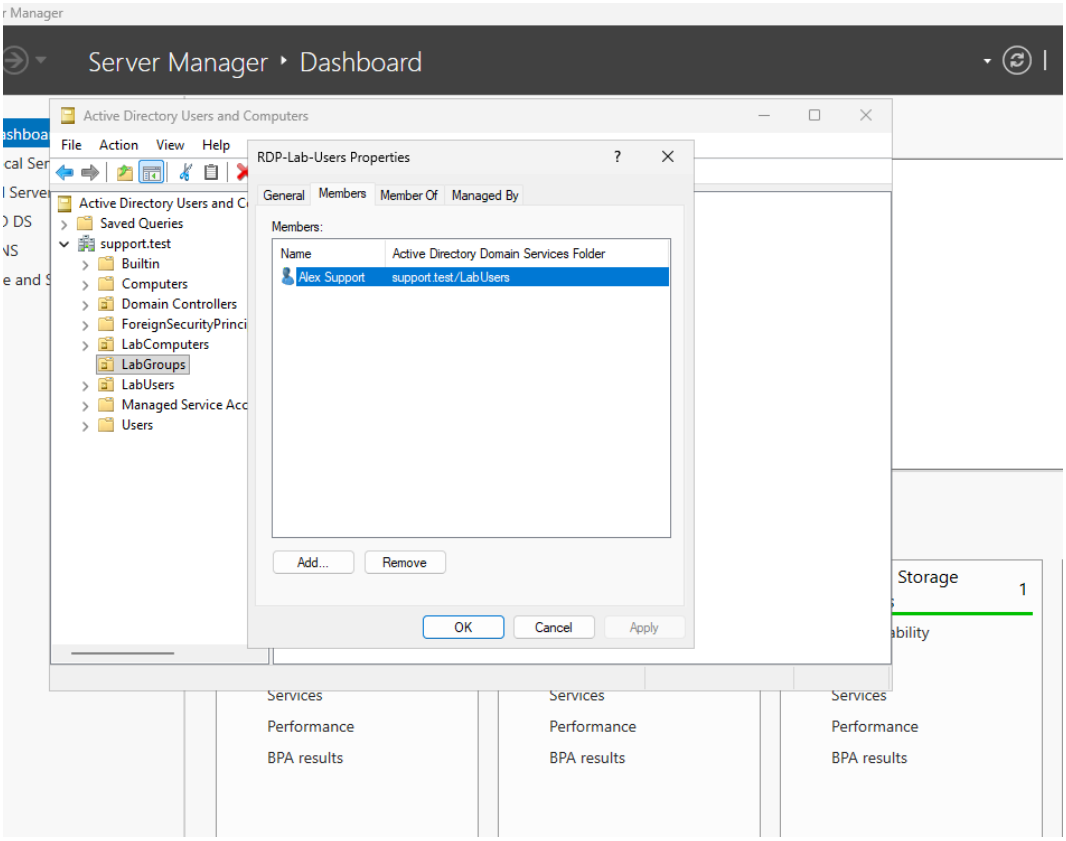
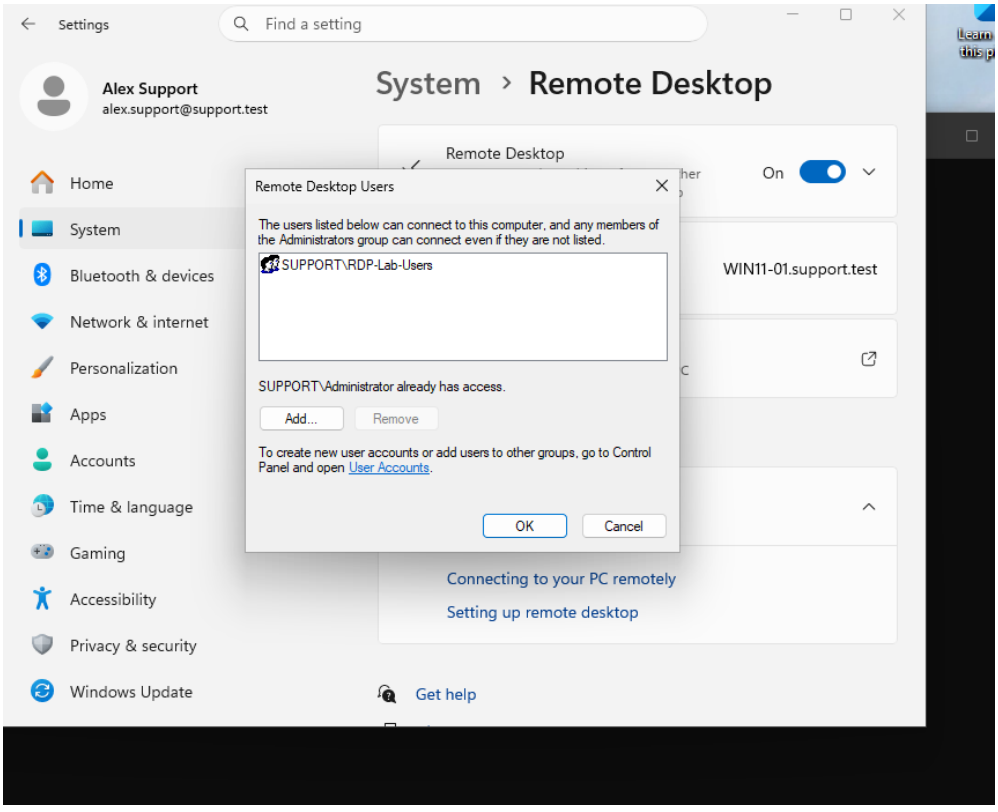


Figure 14. Local Remote Desktop permission (E41). The workstation's local Remote Desktop Users group contains SUPPORT\RDP-Lab-Users.



Appendix A. Cloud license and mailbox access

Figure 15. License assigned to Alex (E47). Alex holds the product license, while the lab administrator is unlicensed. This records service entitlement allocation.

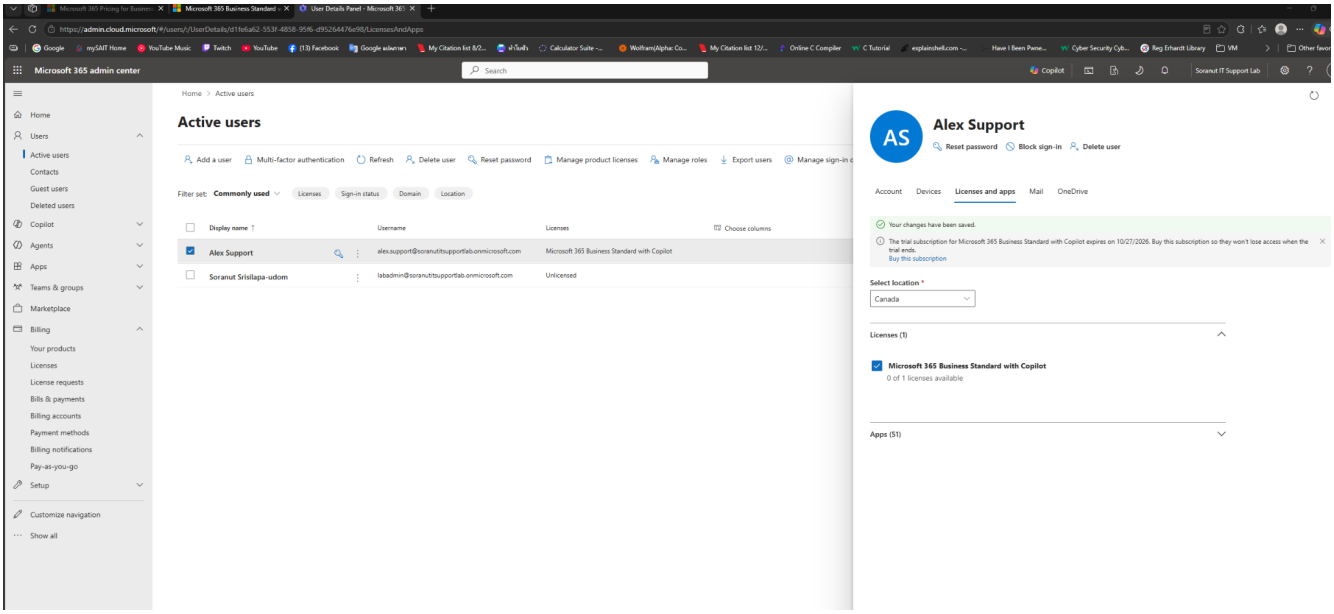
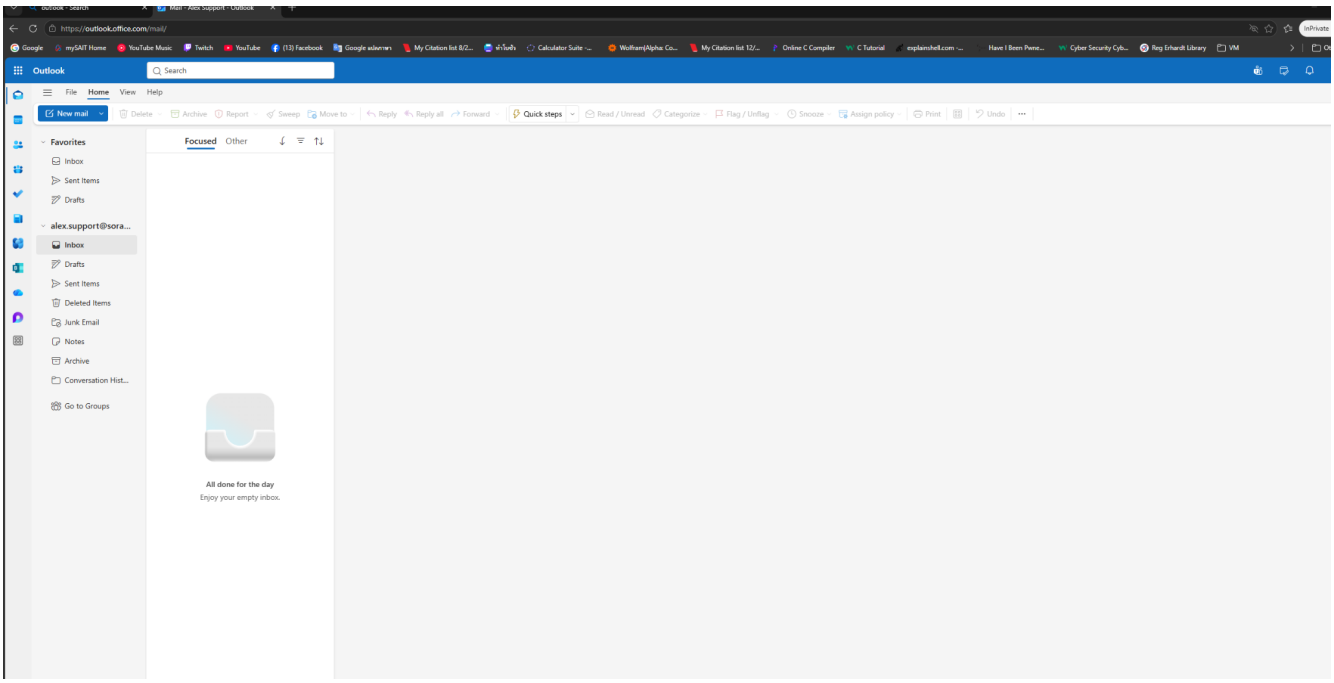


Figure 16. Outlook mailbox access (E51). Outlook opens for the test user. The capture demonstrates mailbox access; mail delivery was not tested separately.



Appendix A. Authentication method and security baseline

Figure 17. Authenticator registration completed (E53). Authenticator Added confirms registration of the authentication method.

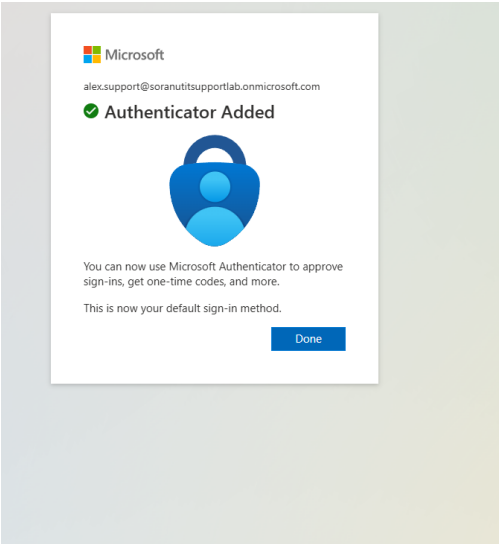
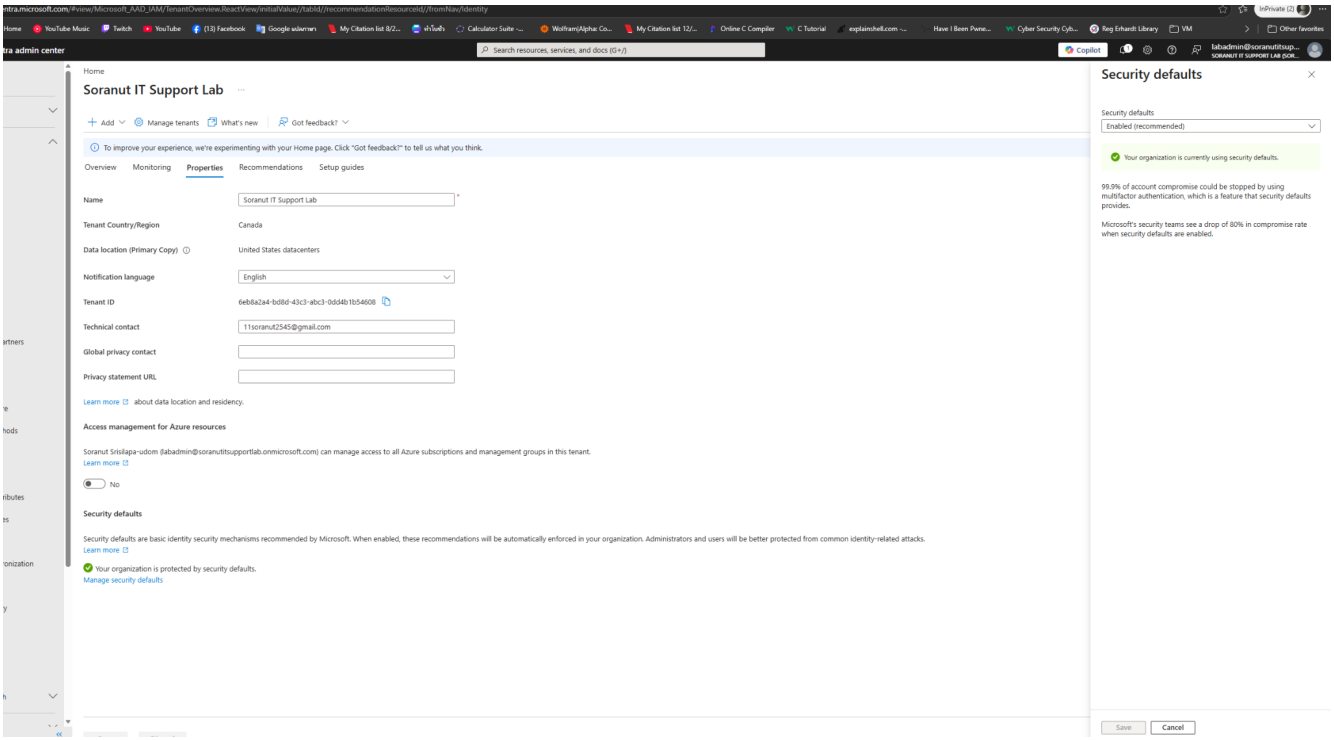
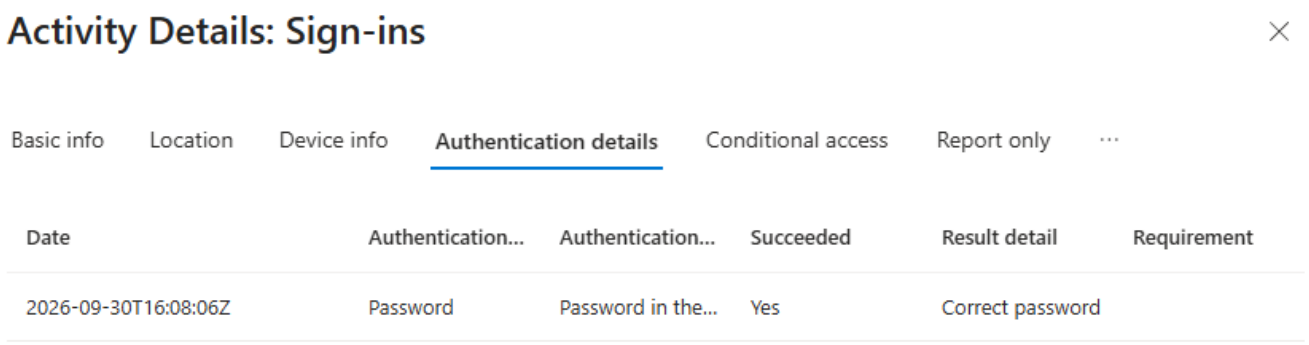


Figure 18. Security Defaults enabled (E55). Security Defaults is enabled in the tenant. This configuration alone does not prove that a particular sign-in used MFA.



Appendix A. Authentication-event interpretation

Figure 19. Authentication details limitation (E61). The inspected sign-in event records Password, succeeded Yes, and Correct password. A separate MFA challenge is not established by this capture.



Interpretation: the event supports a successful password step. Authentication-method enrollment and the tenant baseline are documented separately in Figures 17, 18. A confirmed MFA challenge would require corresponding authentication-detail evidence.

Appendix A. Cloud password reset and access validation

Figure 20. Cloud password reset completed (E62). The administrator receives confirmation that the cloud password was reset. The temporary password is masked.

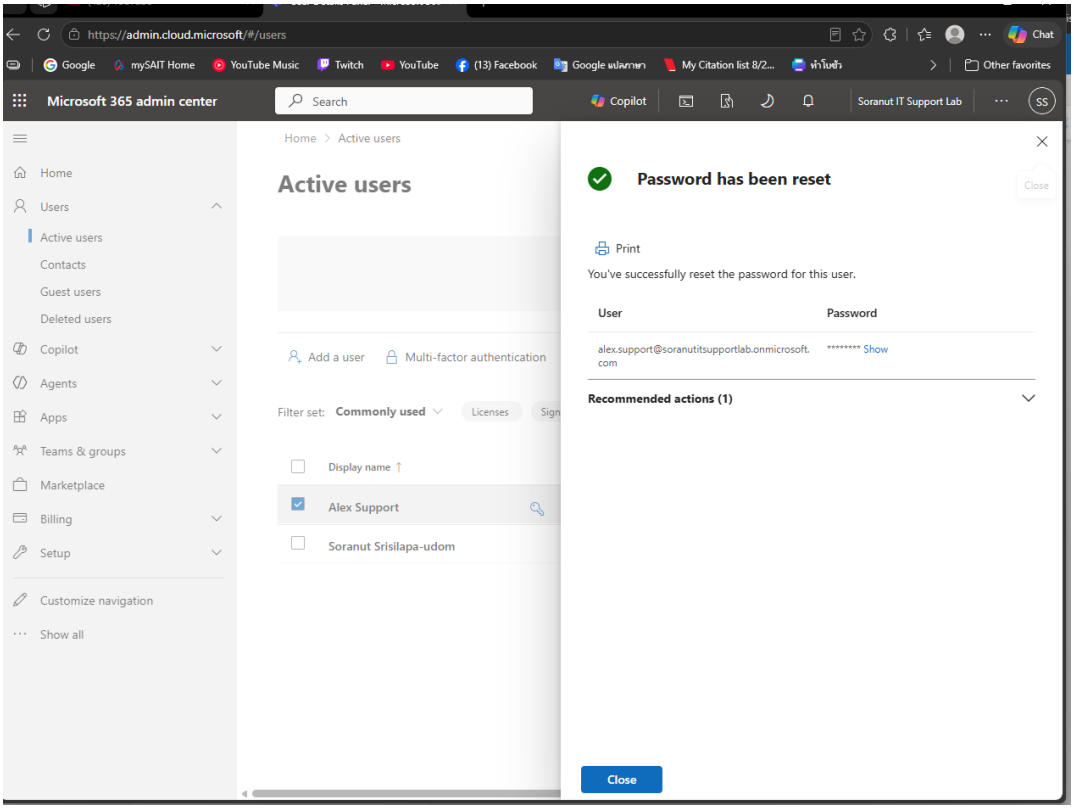
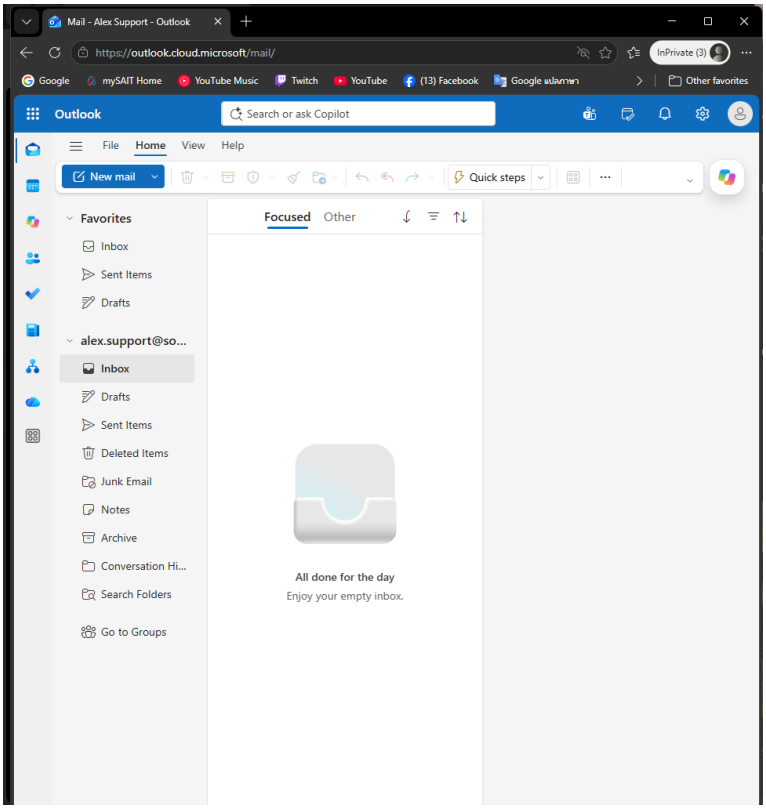


Figure 21. Outlook access after password reset (E64). Outlook opens after the cloud password-change workflow, validating restored service access.



Appendix A. SharePoint access before and after permission

Figure 22. SharePoint access denied before grant (E67). Alex receives You need access before site visitor permission is granted.

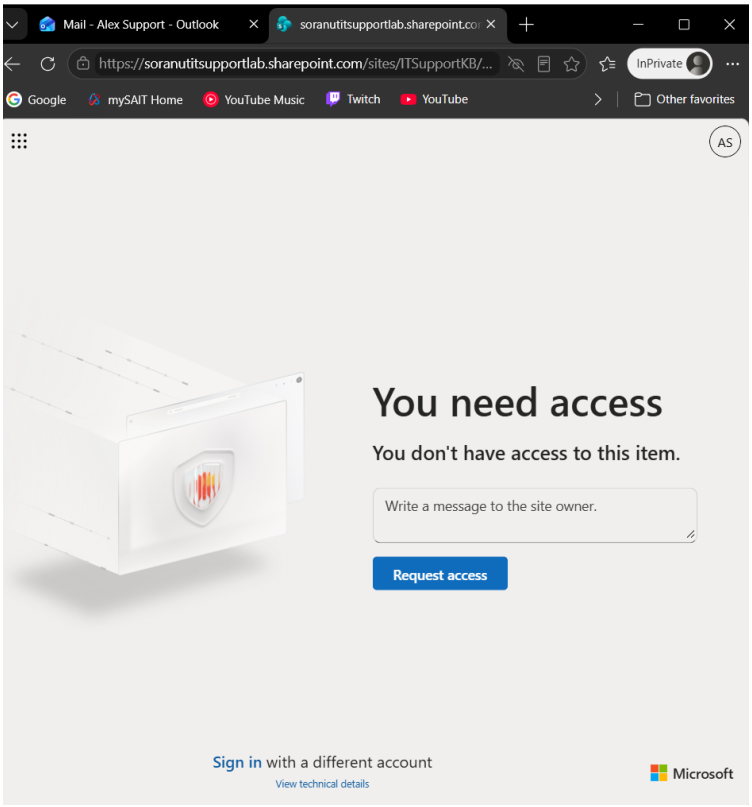
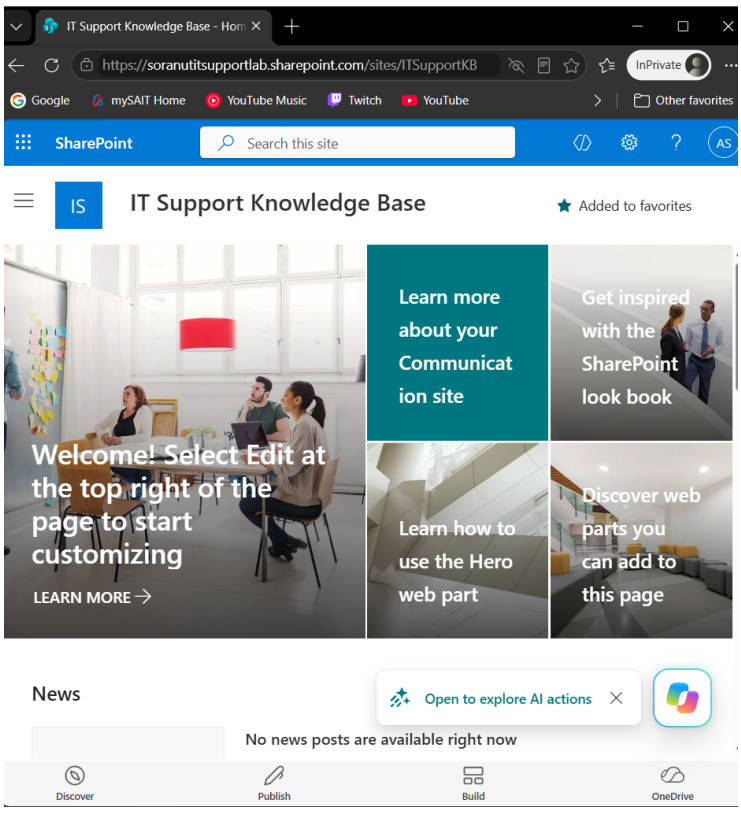


Figure 23. Direct site access verified (E73). The direct SharePoint site URL opens in the test-user context after the visitor grant.



Appendix A. Local account offboarding

Figure 24. AD account disabled (E75). The AD context menu offers Enable Account, indicating that Alex is currently disabled.

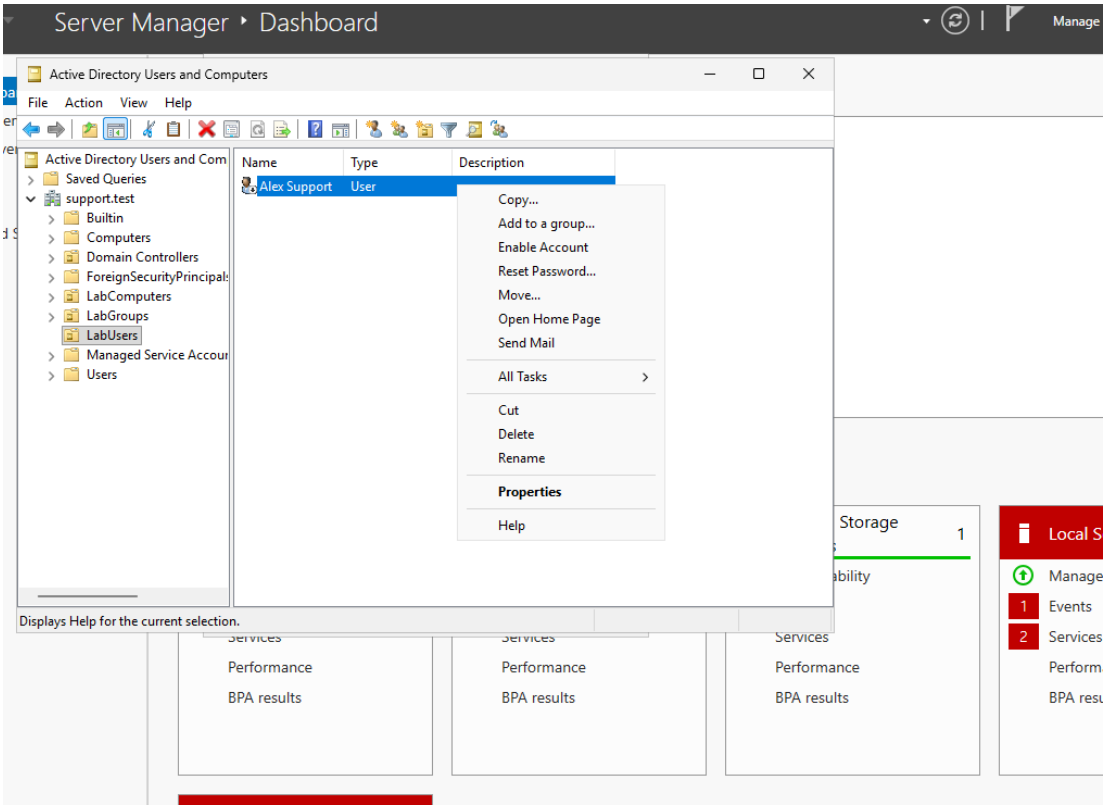
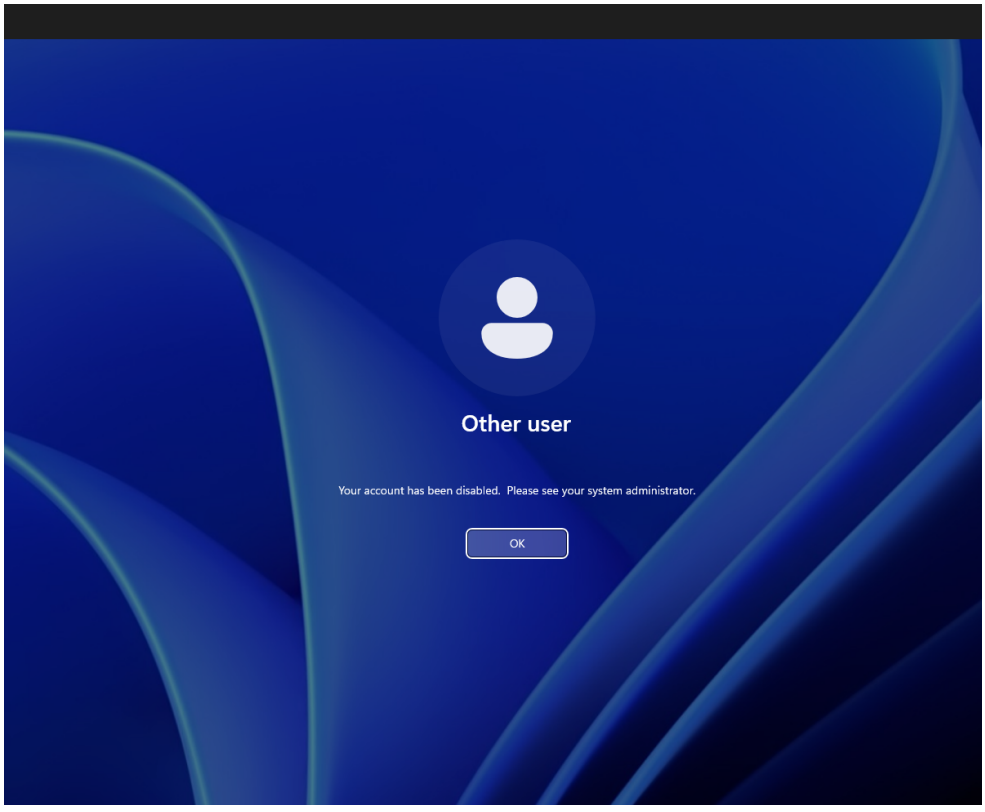


Figure 25. Disabled AD account denied sign-in (E77). A subsequent Windows sign-in attempt is rejected because the AD account is disabled.



Appendix A. Cloud sign-in restriction and session revocation

Figure 26. Microsoft 365 sign-in blocked (E78). The Microsoft 365 administrator view confirms that Alex is blocked from signing in.

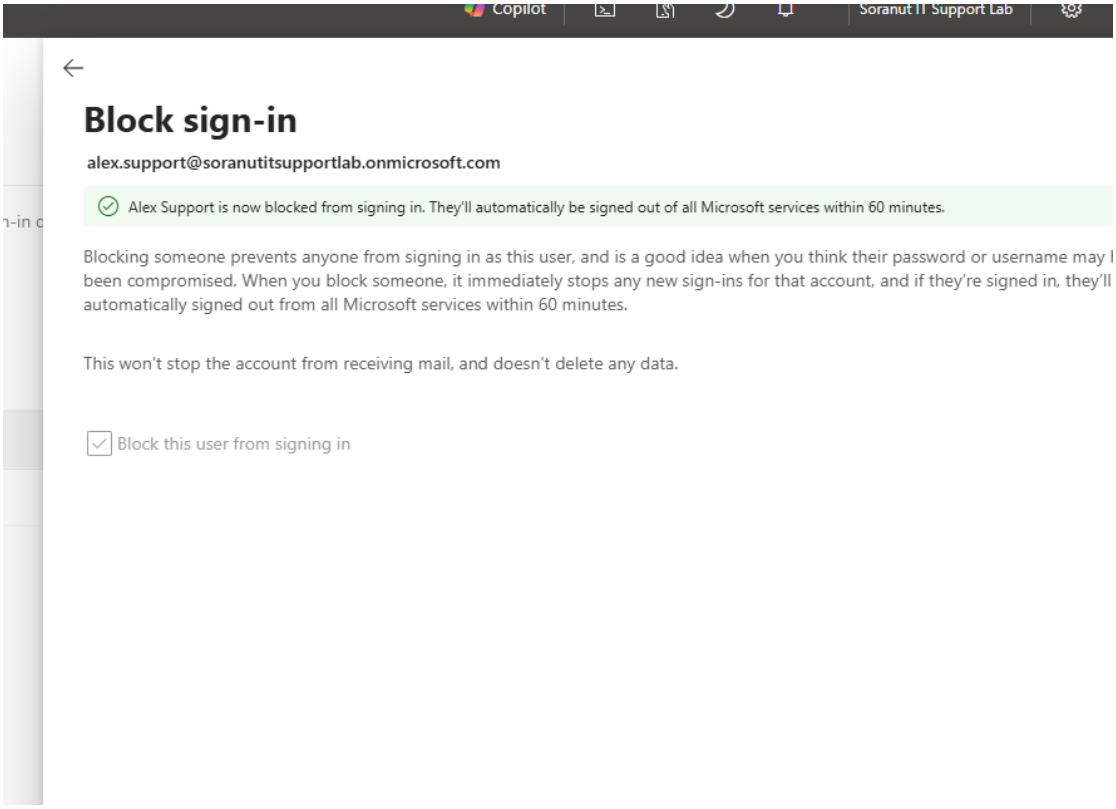
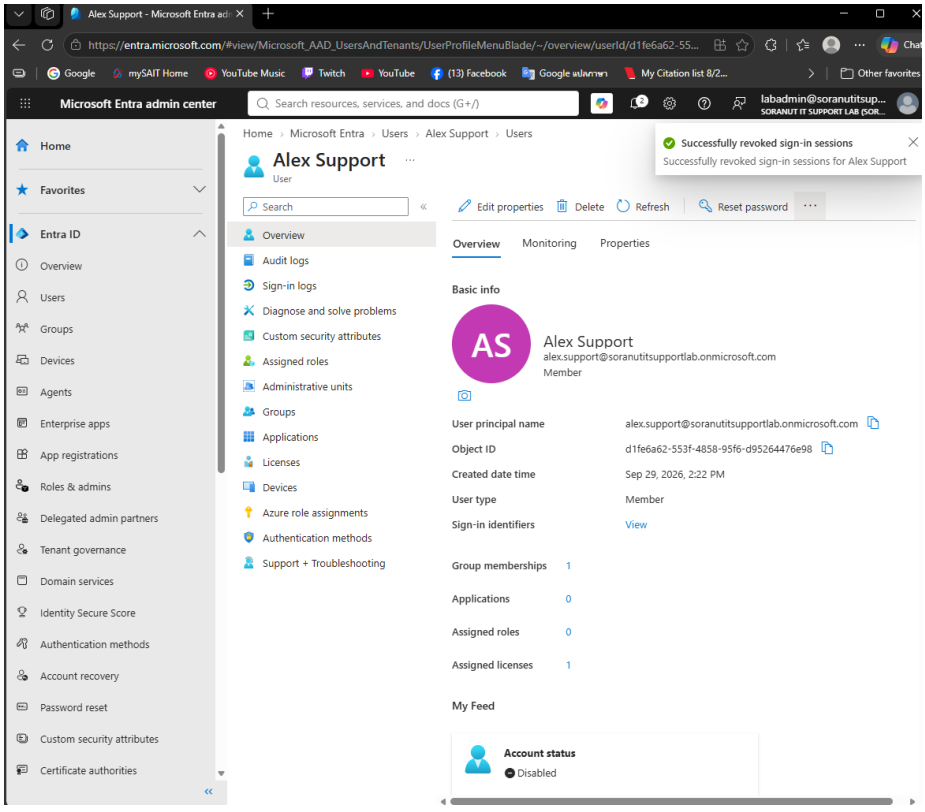


Figure 27. Entra sign-in sessions revoked (E82). Entra confirms that sign-in sessions were revoked. The account is disabled; one license and one group membership remain.



Appendix A. Cloud rejection and SharePoint access removal

Figure 28. Cloud sign-in rejected (E79). A fresh cloud sign-in is rejected after blocking. The screen alone does not identify a specific Entra error code.

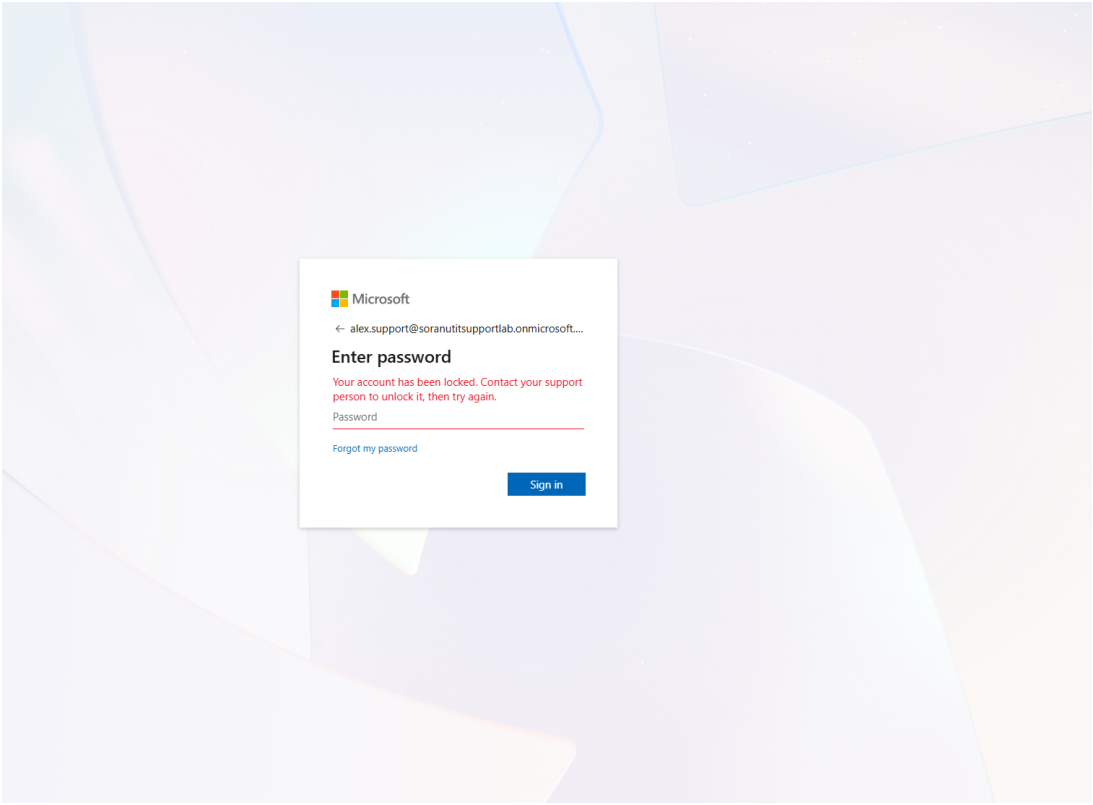


Figure 29. SharePoint visitor permission removed (E84). SharePoint confirms removal of Alex from Site visitors, and the displayed visitor list is empty.

